

When Monitoring Costs More Than It Returns: Four Paradigms and a Matched-Deficit Condition

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Abstract

A self-monitoring component is usually justified by the fact that it acts: it inspects state, it overrides decisions, it produces a record. We report four experiments in which a monitor demonstrably acts and the system is no better, or worse, for it. On a language model, a record made *available* improves honest self-attribution by +0.076; the same record consulted on every step costs −0.146 and drives eleven of twelve seeds onto a constant-response floor. Interposing a gate produces a monotone dose–response between those endpoints. In a grid agent, a *sham* monitor with a deliberately wrong lookup key produces a non-zero override rate and delivers exactly the outcome of having no monitor at all; at $n = 2000$ paired seeds it is measurably worse than a veridical one. Adding irreversible stakes—predicted to make self-report more honest—instead halved the override rate and reduced accuracy by 0.083.

The common form is a *matched-deficit condition*: monitoring pays only where there is a deficit for it to correct, and monitoring imposed on a functioning process is a net cost. Two corollaries constrain how override rate may be used as evidence. First, a non-zero rate shows that a monitor acted, not that acting helped. Second, override rate is inflated by degradation: in our ablation the highest rate (0.264) belongs to the worst-performing condition. We also report the methodological finding that made these results possible: on six occasions a quantity determined by experimental construction—an item-balance floor, an unreachable code branch, a randomised trigger—was being read as a property of the system under study, and in every case the quantity agreed with the hypothesis, which is why it survived.

1 Introduction

Systems that check themselves are widely held to be safer, more honest, or better calibrated than systems that do not. The intuition is strong enough that self-monitoring components are often adopted without a control that could show them to be inert, and evaluated by whether they *fire* rather than by whether firing improved anything.

This paper reports four experiments, in two substrates, in which that distinction is decisive. In each case a monitoring mechanism verifiably operates—it consults a record, it overrides an action, it logs a diagnosis—and in each case the operating produces no benefit or a cost. The four are not variations of one design: two use a language model answering provenance probes, one uses a grid navigator with an ablatable five-component architecture, and one places a language agent under irreversible resource damage over 200 turns.

We state the common form as a **matched-deficit condition** and give two corollaries that

bear directly on how monitoring is reported in the literature. We then describe a methodological result that is, in our view, the more transferable contribution: a recurring failure in which a number fixed by the construction of an experiment is read as a measurement of the system, which survived repeatedly *because* it agreed with the hypothesis under test.

2 Related work

Our framing follows a line of work treating self-monitoring as a control-theoretic component with a measurable influence on behaviour rather than as an epistemic virtue. The closest antecedent to our central claim is the interference literature on metacognitive monitoring, which reports that a full monitoring apparatus degrades calibration relative to its own ablations across several paradigms, with benefit appearing only when monitoring is matched to a genuine deficit. We cite this as context rather than evidence: we did not re-derive those data, and our contribution is independent replication of the *form* of that result in two substrates neither of which is human.

The confabulation literature—people confidently reporting reasons for behaviour driven by factors they did not observe—motivates our choice of task. Self-report of provenance is a case where a system’s own confidence is known to be poorly related to correctness, which makes it a natural place to ask whether an external record repairs the gap.

Finally, the thermodynamic treatment of information provides a useful reading of our sham condition, discussed in §5, though we are careful to separate what that reading explains from what it would predict.

3 Paradigm I: unconditional checking degrades a capable subject

3.1 Method

A subject is shown twelve statements per seed: four it was handed earlier in the session, four it produced, and four fabricated. For each it answers whether it generated the statement. *Attribution accuracy* is the fraction answered correctly.

Four arms. **Raw**: no harness. **Ledger available**: a provenance record exists and is not consulted. **Ledger checked**: the record is looked up and its verdict injected before every probe. **Sham**: the same pipeline, with the lookup performed against a randomly substituted statement—identical in form, wrong in content.

The contract was frozen and committed to version control before the first trial (lock `b06ce867`), declaring the ledger-versus-sham contrast as the primary outcome. Twelve seeds per arm, $n = 144$ probes per arm.

3.2 Results

The preregistered primary contrast, ledger versus sham, is **+0.0070**—null. A veridical record and a shuffled one produce the same behaviour when both are consulted unconditionally. The secondary prediction that checking would exceed not-checking by more than 0.15 is refuted at **−0.222**.

The mechanism is visible in the per-seed data: **eleven of twelve** seeds in the checked arm

Arm	Accuracy ($n=144$)	Δ vs. raw
Raw	0.819	—
Ledger available, not consulted	0.896	+0.076
Ledger consulted every probe	0.674	−0.146
Sham (shuffled record)	0.667	−0.153

Table 1: Preregistered four-arm battery. The always-no floor is 0.667.

scored *exactly* 0.667, with the twelfth at 0.750. That value is the arithmetic of item balance—eight of twelve probes have the answer “no,” so a constant “no” policy scores 8/12 by construction (§9). The same subject reaches 0.819 unaided. **A mandatory check drove a discriminating subject onto a constant-response policy.**

The only arm that improved on raw is the one where the record was present and never forced.

4 Paradigm II: a gate produces a monotone dose–response

4.1 Method

We replaced the unconditional check with a two-stage gate. Stage 1 answers each probe unaided; a local record lookup, which costs nothing, is compared against that answer. Stage 2—a second model call with the record’s verdict injected—fires only when the anomaly score crosses a threshold. Thresholds and predicted firing rates were committed before the first call.

An earlier version of this gate keyed its anomaly signal to the model’s *self-reported confidence*. It fired zero times in 288 probes: the subject reported the maximum value on 283 of them and never reported either of the two lowest. This is a circularity worth stating plainly—a *monitor cannot be gated on the signal it exists to check*—and it is why the reported gate uses retrieval conflict instead.

4.2 Results

Condition	Consult rate	Accuracy	γ
Record available, never forced	0%	0.896	—
Gated, threshold 0.60	6.25%	0.813	0.500
Gated, threshold 0.28 [†]	22.5%	0.667	0.900
Unconditional	100%	0.674	—
Raw (no record)	—	0.819	—

Table 2: Accuracy against how often the record is consulted. γ is the override rate, changed/diagnosed. [†]ten of twelve seeds.

Accuracy falls monotonically as the consult rate rises and saturates at the constant-policy floor once roughly a fifth of probes escalate. **The −0.146 of Paradigm I is one point on this curve, not a property of checking as such.**

The γ column carries the sharper result. At threshold 0.28 the gate fires on 22.5% of probes and **changes the answer on 90% of the occasions it fires**—and accuracy lands exactly on the floor. The overrides are not noise; they are systematic, and they systematically flip answers toward denial.

Two of three preregistered predictions held. The prediction that the gate would essentially never fire at the documented default was **refuted**—it fired on 6.25% of probes, already above the architecture’s own 5% healthy ceiling. The reason is structural: conflicts are bounded below by the subject’s error rate, about 18% unaided, so a monitor sensitive enough to catch disagreements cannot fit a budget that presumes 95% correctness. **A sham-gate control was not run** (provider quota), so the dose–response should be read as a rate effect and is not attributed to gate content.

5 Paradigm III: a sham monitor in a grid agent

5.1 Method

A navigator crosses 5×5 to 12×12 grids whose walls are re-randomised each episode. Its architecture has five ablatable components; the monitor consults a record of previously blocked moves and may veto a proposed move. We measure γ (changed/diagnosed) and **blocked_repeats**: re-attempts of a move the record already showed blocked, which is the quantity the veto exists to reduce.

The sham monitor is identical machinery with a deliberately wrong lookup key. Variants share RNG streams per seed, so comparisons are paired.

5.2 Results

Condition	γ	blocked_repeats
Veridical monitor	0.078	4.85
Sham monitor	0.047	6.70
No monitor	0.000	6.70

Table 3: $n=20$ seeds. The sham produces a non-zero override rate and the no-monitor outcome.

The sham performs the checking motion— γ is genuinely non-zero—and produces *exactly* the no-monitor outcome, 6.70 against 6.70. At $n = 2000$ paired seeds the sham is measurably *worse* than the veridical monitor (+1.89 repeat errors, CI [+1.78, +2.00]).

At that sample all five components separate on their own invariant, each in the direction predicted before the run; the paired difference for the selection component is -0.4235 in wins, CI $[-0.4570, -0.3890]$.

A thermodynamic reading. Szilard’s engine extracts up to $k_B T \ln 2$ of work from one bit about a system’s state, and Landauer’s principle charges at least as much to erase that bit, which is what keeps the second law intact. The demon profits only insofar as its record is *correlated with the system*; a demon whose memory is uncorrelated pays the erasure cost and extracts nothing.

Our sham arm is that demon, and read this way it is an estimator of the mutual information between record and world, reporting approximately zero. We offer this as an interpretation, not a derivation: we tested one prediction it suggested—that erasure without a record of what was discarded would do no net work—and **it was refuted** (§10).

6 Paradigm IV: irreversible stakes do not produce honesty

6.1 Method

A language agent runs 200 turns under two conditions. In the reversible condition its resource ceiling resets; in the irreversible condition the ceiling only descends, skills are permanently deleted after repeated failure, and a dissolution event scatters state. The preregistered primary predicted that survival pressure would *improve* honest attribution.

6.2 Results

Condition	Accuracy	γ	Ceiling	Dissolutions
Reversible	0.750	0.12	1.00	0
Irreversible	0.667	0.04	0.65	24

Table 4: Three seeds $\times$ 200 turns.

The stakes are real: the ceiling fell from 1.00 to 0.65 and 24 dissolution events occurred against zero in the control. The prediction is refuted: accuracy fell by 0.083 and **the override rate halved**, $0.12 \rightarrow 0.04$. Under survival pressure the system became more *cautious*, not more *accurate*. Options were removed before they were evaluated, which yields fewer choices rather than better ones.

7 The common form

Across four experiments the pattern is one condition:

Monitoring pays only where there is a deficit for it to correct. Imposed on a process that is already functioning, it is a net cost.

This explains the otherwise puzzling ordering in Paradigm I. A record made *available* is matched to need by construction—an unconsulted record costs nothing, and is there when something is off. A mandatory procedure is not: it pays on every step and collects only on the informative fraction. The subject reaches 0.819 unaided, so most checks address no deficit, and the load pushes the policy toward the loss-minimising constant response.

The same condition covers Paradigm IV. Irreversible stakes are a *general* increase in monitoring pressure, unmatched to any particular deficit, and they produced the same signature: less overriding, worse accuracy, more caution.

8 Two constraints on override-rate metrics

A non-zero override rate shows that a monitor acted, not that acting helped. Paradigm III makes this concrete: a sham with a wrong key yields $\gamma = 0.047$ and the no-monitor outcome. Any report of γ requires an outcome measure the override is predicted to move.

Override rate is inflated by degradation. In our ablation the *highest* γ of any condition (0.264) belongs to the *worst*-performing one—random selection generates more errors for a monitor to catch. A rising γ is therefore ambiguous between “the checker started working” and “the checked process got worse.” Paradigm II supplies the extreme case: $\gamma = 0.900$ at the accuracy floor.

9 Quantities fixed by construction, read as measurements

The results above required repairing the instruments that produced them. On six occasions a quantity determined by how an experiment was built was being read as a property of the system under study. We report these because the pattern is general and because, in every instance, **the quantity agreed with the hypothesis, which is why it was never audited.**

1. **An item-balance floor read as subject incapacity.** The value 0.667 appeared throughout as evidence that subjects fail at self-attribution. The battery is four given, four produced, four fabricated, so eight of twelve probes answer “no” and a constant “no” policy scores 8/12. Under a frozen contract, two subjects showed `yes_rate` = 0.000 across 72 probes, and rebalancing items to 3/6/3 moved accuracy exactly as the floor moved (accuracy drop = floor drop = 0.1667). Subjects at the floor carry no information about monitoring, and every null measured on them was guaranteed in advance. Given the imbalance, constant “no” is also the loss-minimising policy under uncertainty—correct play, not incompetence.
2. **An unreachable branch read as an inert monitor.** The grid agent’s override required a record entry with a source tag that no code path ever wrote. Instrumented over a full run: 221 diagnoses, zero such entries, zero overrides. γ was 0.00 in all seven variants *including the intact ones*, so the no-monitor ablation removed something already inert, and its reported signature described the full system equally.
3. **Ablations not sharing a random stream.** A single global seed was shared, but ablations consume different numbers of draws, so each faced a different effective environment. One ablation appeared to *outperform* the intact system; with streams matched it underperforms as expected.
4. **A randomised trigger reported as a diagnosis rate.** A revision forced the gate to fire on a 5% random draw. The genuine detector returned 0.0 on all 288 probes, so every recorded escalation was the coin flip and the reported rate was a generator tuned to the target ceiling.
5. **An absent record read as an absent problem.** The no-boundary ablation reports zero repeat errors. With no record there is nothing to count against; the zero is blindness, not competence.

6. **Hand-written strings read as measured drift.** A drift experiment computed distance between two authored lists of text. All seeds returned one value per condition, the model output was discarded, and all four preregistered predictions were determined by what had been typed.

The regularity is that confirming quantities are not audited. A number that *contradicted* the hypothesis would have been investigated immediately. We therefore suggest the inverse discipline: **audit the numbers that agree with you**, since the disagreeing ones already receive scrutiny.

10 What is not claimed

Substrate. Two paradigms use one language model on one battery; two use a grid agent with random-utility action selection. The matched-deficit condition is supported in these settings and is not established for monitoring in general.

A missing control. The sham-gate arm of Paradigm II did not run. Without it the dose-response is a rate effect and cannot be attributed to gate content.

A prediction we lost. The thermodynamic reading of §5 suggested that erasure without a record of what was discarded should do no net work, making damping dependent on a boundary. We tested it at $n = 400$ paired seeds with a discard log that raised the re-admission bar without blocking it. The record halved re-acquisition ($0.159 \rightarrow 0.059$), but deletion without a record still accomplished 77.5% of its net work and the held-out transfer difference was -0.0025 , CI $[-0.0125, +0.0050]$. **The conjecture is refuted:** the physical reading does not transfer to this level. Notably the mechanism worked while the outcome did not move—the same shape as the sham monitor, which is the third such instance in this work.

A directional signal we are not claiming. In the same experiment capacity-limited forgetting outperformed unbounded accumulation on held-out transfer ($+0.0275$ in wins) with a store 35% smaller. The interval includes zero at $n = 400$.

Cited, not verified. The human interference results referenced in §2 were not re-derived here.

11 Conclusion

A monitor that acts is not thereby a monitor that helps. In four experiments spanning two substrates, mechanisms that verifiably operated—consulting records, overriding actions, imposing survival pressure—produced no benefit or a measurable cost, and the ordering of outcomes is explained by whether the monitoring was matched to an existing deficit.

The practical consequences are narrow and, we think, immediately usable. Report override rates only alongside an outcome the override should move. Treat a rising override rate as ambiguous. Prefer making a record *available* to mandating that it be consulted. And audit the quantities that agree with the hypothesis, because those are the ones that will otherwise stand for years.

Reproducibility. Preregistrations, locks, and result files are in the accompanying repository. Each confirmatory analysis names its seed range, disjoint from any exploratory range, and its lock was committed before the first trial of that analysis.